EMPLOYEE ATTRITION DATASET

Simple English Explanation & Project Guide

This document explains the columns in the employee dataset, the types of analysis that can be performed, the most important columns for the project, and the calculated columns that can be created from the existing data.

1. What Does the Dataset Contain?

The dataset contains 19 columns covering employee details, job information, salary, performance, satisfaction, joining dates, and leaving details.

Column	Simple Meaning
Employee_ID	A unique identification number for each employee.
Department	The department where the employee works, such as HR, IT, or Sales.
Job_Role	The employee's job title or role.
Location	The location where the employee works.
Manager_ID	The identification number of the employee's manager.
Employment_Type	The type of employment, such as full-time or contract.
Experience_Years	The employee's total work experience in years.
Age	The employee's age.
Joining_Date	The date the employee joined the company.
Annual_Salary_INR	The employee's annual salary in Indian rupees.
Overtime	Indicates whether the employee works overtime.
Job_Satisfaction_1to4	The employee's job satisfaction score, from 1 to 4.
Performance_Rating_1to4	The employee's performance rating, from 1 to 4.
Employment_Status	The employee's current employment status, such as active or resigned.
Notice_Start_Date	The date the employee's notice period started.
Expected_Last_Working_Day	The expected date of the employee's last working day.
Actual_Last_Working_Day	The actual date the employee left the company.
Attrition	Indicates whether the employee has left the company.
Attrition_Reason	The recorded reason for the employee leaving the company.

2. What Analysis Can Be Performed?

2.1 Employee Attrition Analysis

This analysis helps identify how many employees have left and where attrition is highest.

- How many employees have left the company?
- Which department has the highest attrition?
- Which job roles have more employees leaving?
- What are the main recorded reasons for attrition?

Useful columns: Department, Job_Role, Attrition, Attrition_Reason, and Location.

2.2 Employee Satisfaction and Retention

- Is low job satisfaction associated with higher attrition?
- Are employees who work overtime more likely to leave?
- How are salary, experience, and performance related to attrition?

Useful columns: Job_Satisfaction_1to4, Overtime, Annual_Salary_INR, Experience_Years, and Performance_Rating_1to4.

2.3 Workforce Planning

- How many active employees are currently in the company?
- Which departments may face staff shortages?
- Which employees have an upcoming last working day?
- Which departments may need future hiring?

Useful columns: Department, Employment_Status, Notice_Start_Date, Expected_Last_Working_Day, Actual_Last_Working_Day, and Job_Role.

2.4 Machine Learning – Attrition Prediction

Historical employee data can be used to train a machine learning model to estimate whether an employee may belong to the attrition group.

- Classify attrition as Yes or No.
- Estimate the probability of attrition.
- Identify important factors associated with attrition.

Possible input features include salary, satisfaction, overtime, experience, age, department, performance, and other suitable columns. The target column is Attrition.

3. Which Columns Are Most Important for the Project?

A. Attrition Prediction – Main Machine Learning Task

Target column: Attrition.

Possible input features: Department, Job_Role, Experience_Years, Age, Annual_Salary_INR, Overtime, Job_Satisfaction_1to4, Performance_Rating_1to4, Employment_Type, and Location.

B. Workforce Planning

Important columns: Department, Job_Role, Employment_Status, Joining_Date, Expected_Last_Working_Day, Actual_Last_Working_Day, and Manager_ID.

C. Business Insights and Dashboards

Important columns: Department, Location, Job_Role, Annual_Salary_INR, Attrition_Reason, Overtime, and Job_Satisfaction_1to4.

Important Points

- Employee_ID is mainly used to identify employees. It should generally not be used as a machine learning input feature.
- If Attrition_Reason is recorded only after an employee leaves, it should not be used as an input for attrition prediction. Doing so may cause target leakage.

4. What New Columns Can Be Created from Existing Data?

New calculated columns can be created without changing the original data. These columns can support more detailed analysis.

New Column	How to Create It	Purpose
Tenure_Years	Analysis date – Joining_Date	Shows how many years an employee has worked at the company.
Age_Group	Group employees based on age	Supports age-based attrition analysis.
Experience_Group	Group employees based on experience	Compares freshers with more experienced

		employees.
Salary_Band	Group employees into salary ranges	Supports salary-based attrition comparisons.
Satisfaction_Level	Group satisfaction scores from 1–4 into levels such as Low, Medium, or High	Supports satisfaction analysis.
Performance_Level	Group performance ratings from 1–4	Supports performance comparisons.
Notice_Days	Expected_Last_Working_Day – Notice_Start_Date	Supports notice-period planning.
Days_To_Expected_Exit	Expected_Last_Working_Day – analysis date	Identifies expected upcoming exits.
Attrition_Flag	Yes = 1, No = 0	Creates a numeric target for machine learning classification.
Overtime_Flag	Yes = 1, No = 0	Supports machine learning preprocessing.
Monthly_Salary_INR	Annual_Salary_INR ÷ 12	Supports monthly salary analysis.

Date-based columns depend on the analysis date. If the current date is used, the result may change over time. For historical analysis, using a fixed reporting date is usually better.

5. Can More Advanced Metrics Be Created?

Metric	Meaning / Calculation
Department Attrition Rate	Number of employees leaving ÷ eligible employee population × 100.
Average Tenure at Exit	Average tenure of employees who left the company.
Upcoming Exit Count	Number of employees expected to leave within a specified period.
Department Hiring Gap	Approved target headcount – active headcount.
Attrition Risk Score	The risk probability estimated by a machine learning model.

An attrition risk score is only a prediction. It does not mean that an employee will definitely leave. It should be used alongside a review by the HR team.

6. Simple Project Workflow

1. Understand the dataset columns.
2. Clean the data by checking missing values, duplicates, incorrect formats, and date consistency.
3. Create useful calculated columns.
4. Build attrition and workforce dashboards in Excel or Power BI.
5. Use SQL to identify insights related to departments, salary, satisfaction, and attrition.
6. Build a machine learning model to predict Attrition as Yes or No.
7. Evaluate the results and present them clearly to HR, including the model's limitations.

7. How Is Machine Learning Used in This Project?

7.1 Main Purpose of Machine Learning

For example, assume a company has 1,000 employees and historical records of employees who have left. The records may include salary, job satisfaction, overtime, and work experience.

- The machine learning model learns patterns from historical employee data.
- The model uses these patterns to analyze suitable new employee records.
- It estimates the probability that a new record may belong to the attrition group.

In simple terms, the machine learning task is to learn patterns from past employee data and use them to estimate future attrition risk.